

REU-HPC: Week-6 Group Meeting

Group Meetings

Tuesday 3-4pm (ROC room 1.109)

<https://utexas.zoom.us/j/85978334209>

Week 6 Objectives (July 6 – July 12)

The primary goal for week 6:

- Run sample workloads with and without PEAK to collect profiling data and analyze the usage of mathematical libraries such as BLAS, LAPACK, and FFTW.
- Explore and create build scripts for more software packages from the Top 100 applications list by reviewing official documentation, searching for existing examples, and leveraging AI-assisted guidance.
- Meeting with HPC professionals: I'm going to arrange more meetings with members of the HPC group so you can learn more about their work at TACC and hear about their experiences in the field of high-performance computing.

Useful information

- **New PEAK, readme, and documentation:**
Link: <https://github.com/peak-team/peak>
- Project directory: `/work2/05392/cylu/share/reu_2026/2.project`
 - **App build scripts:** abinit, berkeley_gw, dftb+, epw, gpaw, gromacs, lammps, lapack, meep, milc, openblas, yambo
 - **Examples:** cp2k, dftb+, gromacs, lammps, milc, ml, siesta, yambo
- **Frontera-top100-2025-1.txt:**
Link:
<https://drive.google.com/file/d/15VXVPnREFcz3hiVyVoCW8ThX5TO6WELO/view?usp=sharing>
- **package_list.xlsx**
Link:
<https://docs.google.com/spreadsheets/d/1JSHFZpOV8IA3M1mw86WJAUWZ8kotStPf/edit?usp=sharing&oid=101096215360805228479&rtpof=true&sd=true>